

```

/* =====
COMPANY DATABASE STORED PROCEDURE LAB
Beginner Level MSSQL Stored Procedure Examples
SQL Server Version
===== */

-- =====
-- STEP 1: CREATE DATABASE
-- =====

CREATE DATABASE company_db;
GO

USE company_db;
GO

-- =====
-- STEP 2: CREATE EMPLOYEES TABLE
-- =====

CREATE TABLE employees (
    employee_id INT IDENTITY(1,1) PRIMARY KEY,
    employee_name VARCHAR(100) NOT NULL,
    department VARCHAR(100),
    salary DECIMAL(10,2),
    address VARCHAR(255),
    phone_number VARCHAR(20),
    created_at DATETIME DEFAULT GETDATE()
);
GO

-- =====
-- STEP 3: INSERT SAMPLE DATA
-- =====

INSERT INTO employees
(employee_name, department, salary, address, phone_number)
VALUES

```

```

('Aung Aung', 'IT', 800000, 'Yangon', '091111111'),
('Su Su', 'HR', 600000, 'Mandalay', '092222222'),
('Kyaw Kyaw', 'Finance', 900000, 'Naypyitaw', '093333333'),
('Mya Mya', 'Marketing', 700000, 'Yangon', '094444444'),
('Ko Ko', 'IT', 850000, 'Bago', '095555555');
GO

```

```

-- =====
-- VIEW CURRENT EMPLOYEE DATA
-- =====

```

```

58  SELECT * FROM employees;
59  GO
60

```

100 % 2 0

Results Messages

	employee_id	employee_name	department	salary	address	phone_number	created_at
1	1	Aung Aung	IT	800000.00	Yangon	091111111	2026-05-20 13:11:52.927
2	2	Su Su	HR	600000.00	Mandalay	092222222	2026-05-20 13:11:52.927
3	3	Kyaw Kyaw	Finance	900000.00	Naypyitaw	093333333	2026-05-20 13:11:52.927
4	4	Mya Mya	Marketing	700000.00	Yangon	094444444	2026-05-20 13:11:52.927
5	5	Ko Ko	IT	850000.00	Bago	095555555	2026-05-20 13:11:52.927

```

/* =====
STORED PROCEDURE 1:
InsertEmployee
PURPOSE:
Insert new employee records
===== */

```

```

CREATE PROCEDURE InsertEmployee
    @p_employee_name VARCHAR(100),
    @p_department VARCHAR(100),
    @p_salary DECIMAL(10,2),
    @p_address VARCHAR(255),
    @p_phone_number VARCHAR(20)
AS
BEGIN

```

```

    INSERT INTO employees(
        employee_name,
        department,

```

```

        salary,
        address,
        phone_number
    )
VALUES(
    @p_employee_name,
    @p_department,
    @p_salary,
    @p_address,
    @p_phone_number
);

END;
GO

-- =====
-- EXECUTE InsertEmployee PROCEDURE
-- =====

EXEC InsertEmployee
    'Hla Hla',
    'Admin',
    500000,
    'Yangon',
    '096666666';
GO

EXEC InsertEmployee
    'Hla Hla Moe',
    'IT',
    900000,
    'Taung Gyi',
    '096666667';
GO

```

-- =====

-- VIEW DATA AFTER INSERT

-- =====

```
137 SELECT * FROM employees;
```

```
138 GO
```

100 %

✖ 2

⚠ 0

↑ ↓

Results

Messages

	employee_id	employee_name	department	salary	address	phone_number	created_at
1	1	Aung Aung	IT	800000.00	Yangon	091111111	2026-05-20 13:11:52.927
2	2	Su Su	HR	600000.00	Mandalay	092222222	2026-05-20 13:11:52.927
3	3	Kyaw Kyaw	Finance	900000.00	Naypyitaw	093333333	2026-05-20 13:11:52.927
4	4	Mya Mya	Marketing	700000.00	Yangon	094444444	2026-05-20 13:11:52.927
5	5	Ko Ko	IT	850000.00	Bago	095555555	2026-05-20 13:11:52.927
6	6	Hla Hla	Admin	500000.00	Yangon	096666666	2026-05-20 13:12:58.273
7	7	Hla Hla Moe	IT	900000.00	Taung Gyi	096666667	2026-05-20 13:13:02.120

/* =====

STORED PROCEDURE 2:

DeleteEmployee

PURPOSE:

Delete employee using employee ID

===== */

CREATE PROCEDURE DeleteEmployee

@p_employee_id INT

AS

BEGIN

DELETE FROM employees

WHERE employee_id = @p_employee_id;

END;

GO

-- =====

-- EXECUTE DeleteEmployee PROCEDURE

-- =====

EXEC DeleteEmployee 6;

GO

-- =====

-- VIEW DATA AFTER DELETE

-- =====

```
171  SELECT * FROM employees;  
172  GO  
173
```

100 %  2  0  

 Results  Messages

	employee_id	employee_name	department	salary	address	phone_number	created_at
1	1	Aung Aung	IT	800000.00	Yangon	091111111	2026-05-20 13:11:52.927
2	2	Su Su	HR	600000.00	Mandalay	092222222	2026-05-20 13:11:52.927
3	3	Kyaw Kyaw	Finance	900000.00	Naypyitaw	093333333	2026-05-20 13:11:52.927
4	4	Mya Mya	Marketing	700000.00	Yangon	094444444	2026-05-20 13:11:52.927
5	5	Ko Ko	IT	850000.00	Bago	095555555	2026-05-20 13:11:52.927
6	7	Hla Hla Moe	IT	900000.00	Taung Gyi	096666667	2026-05-20 13:13:02.120

/* =====

STORED PROCEDURE 3:

UpdateEmployeeAddress

PURPOSE:

Update employee address

===== */

CREATE PROCEDURE UpdateEmployeeAddress

@p_employee_id INT,

@p_new_address VARCHAR(255)

AS

BEGIN

UPDATE employees

SET address = @p_new_address

WHERE employee_id = @p_employee_id;

END;

GO

-- =====

-- EXECUTE UpdateEmployeeAddress PROCEDURE

-- =====

EXEC UpdateEmployeeAddress 2, 'Taunggyi';

GO

-- =====

-- VIEW DATA AFTER UPDATE

-- =====

```
210  SELECT * FROM employees;
211  GO
```

100 % 2 0 ↑ ↓

Results Messages

	employee_id	employee_name	department	salary	address	phone_number	created_at
1	1	Aung Aung	IT	800000.00	Yangon	091111111	2026-05-20 13:11:52.927
2	2	Su Su	HR	600000.00	Taunggyi	092222222	2026-05-20 13:11:52.927
3	3	Kyaw Kyaw	Finance	900000.00	Naypyitaw	093333333	2026-05-20 13:11:52.927
4	4	Mya Mya	Marketing	700000.00	Yangon	094444444	2026-05-20 13:11:52.927
5	5	Ko Ko	IT	850000.00	Bago	095555555	2026-05-20 13:11:52.927
6	7	Hla Hla Moe	IT	900000.00	Taung Gyi	096666667	2026-05-20 13:13:02.120

/* =====

STORED PROCEDURE 4:

GetEmployees

PURPOSE:

Retrieve all employee records

===== */

CREATE PROCEDURE GetEmployees

AS

BEGIN

SELECT *

FROM employees;

END;

GO

-- =====

-- EXECUTE GetEmployees PROCEDURE

-- =====

```
235 EXEC GetEmployees;
```

```
236 GO
```

```
237
```

100 %  2  0  

 Results  Messages

	employee_id	employee_name	department	salary	address	phone_number	created_at
1	1	Aung Aung	IT	800000.00	Yangon	091111111	2026-05-20 13:11:52.927
2	2	Su Su	HR	600000.00	Taunggyi	092222222	2026-05-20 13:11:52.927
3	3	Kyaw Kyaw	Finance	900000.00	Naypyitaw	093333333	2026-05-20 13:11:52.927
4	4	Mya Mya	Marketing	700000.00	Yangon	094444444	2026-05-20 13:11:52.927
5	5	Ko Ko	IT	850000.00	Bago	095555555	2026-05-20 13:11:52.927
6	7	Hla Hla Moe	IT	900000.00	Taung Gyi	096666667	2026-05-20 13:13:02.120

/* =====

STORED PROCEDURE 5:

GetEmployeeCount

PURPOSE:

Return total employee count

OUTPUT PARAMETER EXAMPLE

===== */

CREATE PROCEDURE GetEmployeeCount

@p_total_employees INT OUTPUT

AS

BEGIN

SELECT @p_total_employees = COUNT(*)

FROM employees;

END;

GO

```

-- =====
-- EXECUTE GetEmployeeCount PROCEDURE
-- =====

263     DECLARE @total INT;
264
265     EXEC GetEmployeeCount @total OUTPUT;
266
267     SELECT @total AS total_employees;
268     GO

```

100 % 2 0 ↑ ↓

Results Messages

	total_employees
1	6

```

/* =====
STORED PROCEDURE 6:
IncreaseSalary
PURPOSE:
Increase employee salary and
return updated salary
===== */

```

```

CREATE PROCEDURE IncreaseSalary
    @p_employee_id INT,
    @p_increment DECIMAL(10,2) ,
    @p_newsal DECIMAL(10,2) OUTPUT
AS
BEGIN

    -- Increase salary
    UPDATE employees
    SET salary = salary + @p_increment
    WHERE employee_id = @p_employee_id;
    -- Return updated salary
    SELECT @p_newsal = salary
    FROM employees
    WHERE employee_id = @p_employee_id;

END;
GO

```

-- =====

-- EXECUTE IncreaseSalary PROCEDURE

-- =====

```
311 DECLARE @inc_salary DECIMAL(10,2);
312 DECLARE @new_salary DECIMAL(10,2);
313
314 SET @inc_salary = 50000;
315
316 EXEC IncreaseSalary 1, @inc_salary,@new_salary OUTPUT;
317
318 SELECT @new_salary AS updated_salary;
319 GO
320
```

100 % No issues found

Results Messages

	updated_salary
1	850000.00

-- =====

-- VIEW ALL STORED PROCEDURES

-- =====

```
325 SELECT
326     name,
327     create_date,
328     modify_date
329 FROM sys.procedures;
330 GO
331
```

100 % No issues found

Results Messages

	name	create_date	modify_date
1	InsertEmployee	2026-05-20 13:12:52.097	2026-05-20 13:12:52.097
2	DeleteEmployee	2026-05-20 13:16:39.160	2026-05-20 13:16:39.160
3	UpdateEmployeeAddress	2026-05-20 13:18:06.523	2026-05-20 13:18:06.523
4	GetEmployees	2026-05-20 13:19:21.063	2026-05-20 13:19:21.063
5	GetEmployeeCount	2026-05-20 13:21:37.640	2026-05-20 13:21:37.640
6	IncreaseSalary	2026-05-20 13:27:01.187	2026-05-20 13:27:01.187

-- =====

-- VIEW PROCEDURE DEFINITION

-- =====

```
336 EXEC sp_helptext 'GetEmployees';
337 GO
338
```

100 % No issues found

	Text
1	CREATE PROCEDURE GetEmployees
2	
3	AS
4	BEGIN
5	
6	SELECT *
7	FROM employees;
8	
9	END;

-- =====

-- ALTER STORED PROCEDURE EXAMPLE

-- =====

ALTER PROCEDURE GetEmployees

AS

BEGIN

SELECT

employee_id,

employee_name,

department,

salary

FROM employees;

END;

GO

```
-- =====  
-- EXECUTE ALTERED PROCEDURE  
-- =====
```

```
362 EXEC GetEmployees;  
363 GO
```

100 % No issues found

Results Messages

	employee_id	employee_name	department	salary
1	1	Aung Aung	IT	850000.00
2	2	Su Su	HR	600000.00
3	3	Kyaw Kyaw	Finance	900000.00
4	4	Mya Mya	Marketing	700000.00
5	5	Ko Ko	IT	850000.00
6	7	Hla Hla Moe	IT	900000.00

```
-- Local temporary stored procedures
```

```
DROP PROCEDURE IF EXISTS #TempEmployeeReport;  
GO
```

```
CREATE PROCEDURE #TempEmployeeReport  
AS  
BEGIN
```

```
    SELECT  
        employee_id,  
        employee_name,  
        department,  
        salary  
    FROM employees  
    WHERE salary >= 700000;
```

```
END;  
GO
```

385 EXEC #TempEmployeeReport;
 386 GO
 387

100 % No issues found

Results Messages

	employee_id	employee_name	department	salary
1	1	Aung Aung	IT	850000.00
2	3	Kyaw Kyaw	Finance	900000.00
3	4	Mya Mya	Marketing	700000.00
4	5	Ko Ko	IT	850000.00
5	7	Hla Hla Moe	IT	900000.00

SQLQuery1.s... Yu Ya (85))* X StoredProced...y Yu Y

```

1  USE company_db;
2  GO
3
4  EXEC #TempEmployeeReport;
5  GO
  
```

100 % 1 0

Messages

Msg 2812, Level 16, State 62, Line 4
 Could not find stored procedure '#TempEmployeeReport'.

Completion time: 2026-05-20T13:34:28.2682622+06:30

-- Global temporary stored procedures

```

DROP PROCEDURE IF EXISTS ##GlobalEmployeeReport;
GO
  
```

```

CREATE PROCEDURE ##GlobalEmployeeReport
AS
BEGIN
  
```

```

    SELECT
      employee_id,
      employee_name,
      department,
      salary
    FROM employees
    WHERE salary >= 800000;
  
```

```

END;
GO
  
```

```

408 EXEC ##GlobalEmployeeReport;
409 GO

```

100 % No issues found

Results Messages

	employee_id	employee_name	department	salary
1	1	Aung Aung	IT	850000.00
2	3	Kyaw Kyaw	Finance	900000.00
3	5	Ko Ko	IT	850000.00
4	7	Hla Hla Moe	IT	900000.00

SQLQuery1.s... Yu Ya (85))* StoredProced...y

```

1 USE company_db;
2 GO
3
4 EXEC ##GlobalEmployeeReport;
5 GO
6

```

100 % 1 0

Results Messages

	employee_id	employee_name	department	salary
1	1	Aung Aung	IT	850000.00
2	3	Kyaw Kyaw	Finance	900000.00
3	5	Ko Ko	IT	850000.00
4	7	Hla Hla Moe	IT	900000.00

-- Check Session IDs

```

413 SELECT @@SPID AS session_id;

```

100 % No issues found

Results Messages

	session_id
1	95

Different Session

The screenshot displays two SQL query windows. The top window, titled 'SQLQuery1.s... Yu Ya (53))*', shows a successful query execution. The query is 'SELECT @@spid AS session_id; GO'. The results pane shows a single row with 'session_id' as 53. The bottom window, titled 'SQLQuery1.s... Yu Ya (53))*', shows an error. The query is 'EXEC ##GlobalEmployeeReport; GO'. The error message is 'Msg 2812, Level 16, State 62, Line 4: Could not find stored procedure '##GlobalEmployeeReport'.'. The completion time is '2026-05-20T13:52:05.1389039+06:30'.

```
-- =====
-- DROP PROCEDURE EXAMPLES
-- =====

-- DROP PROCEDURE InsertEmployee;
-- DROP PROCEDURE DeleteEmployee;
-- DROP PROCEDURE UpdateEmployeeAddress;
-- DROP PROCEDURE GetEmployees;
-- DROP PROCEDURE GetEmployeeCount;
-- DROP PROCEDURE IncreaseSalary;

-- =====
-- END OF MSSQL LAB SCRIPT
-- =====
```